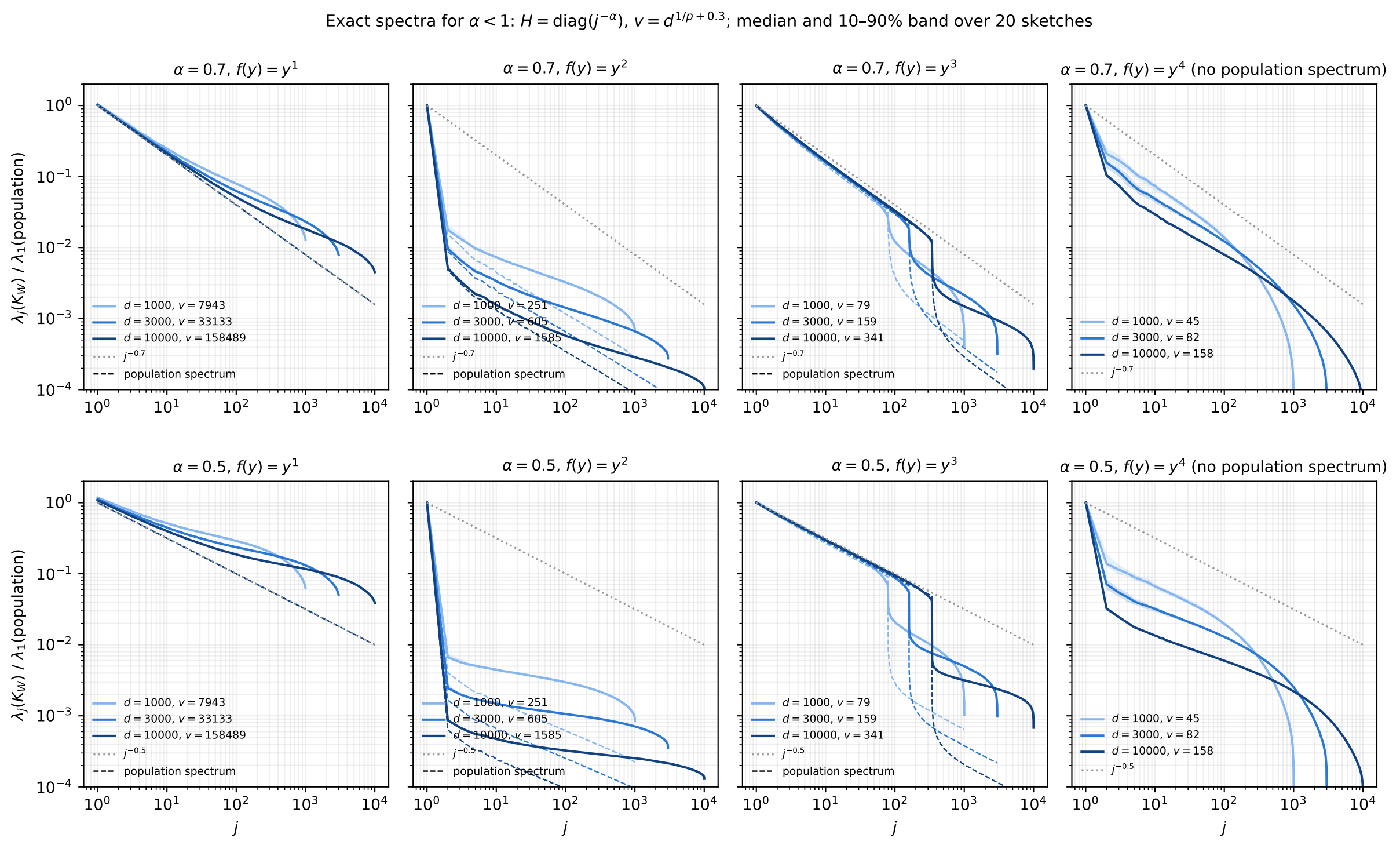

Exact spectra for $\alpha < 1$: $H = \mathrm{diag}(j^{-\alpha})$, $v = d^{1/p+0.3}$; median and 10–90% band over 20 sketches
$\alpha = 0.7$, $f(y) = y^1$
$\alpha = 0.7$, $f(y) = y^2$
$\alpha = 0.7$, $f(y) = y^3$
$\alpha = 0.7$, $f(y) = y^4$ (no population spectrum)
$\alpha = 0.5$, $f(y) = y^1$
$\alpha = 0.5$, $f(y) = y^2$
$\alpha = 0.5$, $f(y) = y^3$
$\alpha = 0.5$, $f(y) = y^4$ (no population spectrum)
$\lambda_j(K_W) / \lambda_1$(population)
j
$d = 1000$, $v = 7943$
$d = 3000$, $v = 33133$
$d = 10000$, $v = 158489$
$d = 1000$, $v = 251$
$d = 3000$, $v = 605$
$d = 10000$, $v = 1585$
$d = 1000$, $v = 79$
$d = 3000$, $v = 159$
$d = 10000$, $v = 341$
$d = 1000$, $v = 45$
$d = 3000$, $v = 82$
$d = 10000$, $v = 158$
$j^{-0.7}$
$j^{-0.5}$
population spectrum